

## **SIMATS ENGINEERING**

### **ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1714

#### **COURSE OUTCOME COVERED**

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration: 1 Hour

QUESTIONS: 4 Total: Marks:40

#### **Annexure A**

##### **Code of Conduct:**

I ***D Dharshana(192524243)*** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

*D Dharshana*

**Signature of the student:**

## **INDUSTRY PROBLEM TASK**

### **INDUSTRY SCENARIO**

**Context:** A manufacturing plant has collected sensor records (vibration level, temperature, operating hours, past maintenance count, load percentage) to build a predictive model for identifying machines likely to fail. The plant also wants to train an AI agent to recommend maintenance actions (Inspect / Lubricate / Replace Part / Continue Operation) based on a machine's condition over time. You are hired as an AI/ML Engineer. Address the following tasks:

#### **Task 1: Data Preparation & Inductive Learning**

- (a) Describe the inductive learning approach suitable for this manufacturing dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (far fewer failure events than normal operation records) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a predictive-maintenance use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

#### **Task 2: Decision Tree for Equipment Failure Prediction**

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed failures) and suggest how to reduce them - justify from a safety perspective.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment on the factory floor.

#### **Task 3: Statistical Learning for Risk Stratification**

(a) Apply Naive Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of failure from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

#### **Task 4: Reinforcement Learning for Action Recommendation**

(a) Model the maintenance recommendation as an MDP. Define: States (machine health stages), Actions (Inspect / Lubricate / Replace Part / Continue Operation), Reward function (uptime improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 machine episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for maintenance decisions (worker safety, cost trade-offs, explainability).

### **INDUSTRY PROBLEM TASK**

#### **Predictive Maintenance and Reinforcement Learning for Manufacturing Equipment**

##### **TASK 1 – DATA PREPARATION & INDUCTIVE LEARNING**

###### **1(a) Inductive Learning Approach**

Inductive learning learns a general relationship from previously observed examples and uses it to make predictions on new data. For this manufacturing problem, historical machine records containing sensor measurements and failure outcomes can be used to train a supervised machine-learning model.

The target variable is:

###### **Machine Failure**

- 0 – Normal operation
- 1 – Machine failure

Important features include:

| Feature                | Justification                                                                                                                |
|------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Vibration Level        | Increasing vibration may indicate mechanical wear, imbalance, or bearing problems and can be an early indication of failure. |
| Temperature            | Abnormally high temperature can indicate overheating, friction, or cooling-system problems.                                  |
| Operating Hours        | Machines with more operating hours may experience greater wear and have a higher probability of failure.                     |
| Past Maintenance Count | A machine with frequent previous maintenance may have recurring equipment problems.                                          |
| Load Percentage        | Continuous operation at high load can increase stress and temperature and therefore increase failure risk.                   |

The learning process is:

**Historical sensor data → Preprocessing → Model training → Failure prediction → Maintenance decision**

### 1(b) Handling Class Imbalance

In predictive maintenance, failure events are normally much fewer than normal-operation records. For example, the dataset may contain 950 normal records and only 50 failure records.

This creates a **class imbalance problem**. If the model is trained directly on this dataset, it may become biased toward predicting the majority class.

**Suitable Strategy: SMOTE**

**SMOTE (Synthetic Minority Over-sampling Technique)** can be used to create synthetic examples of the minority failure class.

SMOTE should be applied **only to the training set**, after splitting the data, so that information from the test set does not leak into the training process.

### **1(c) Train-Test Split and Preprocessing**

The dataset is divided into:

- **70% Training Data**
- **30% Testing Data**

The 70:30 ratio provides sufficient data for learning while keeping a large enough independent test set for evaluating predictive performance.

#### **Preprocessing Steps**

1. **Missing values:** Missing sensor values are identified and replaced using suitable methods such as median imputation.
2. **Normalisation:** Numerical features such as temperature, vibration and load percentage are scaled to a common range.
3. **Encoding:** Categorical variables, if present, are converted into numerical form using techniques such as one-hot encoding.
4. **Class balancing:** SMOTE is applied to the training data.
5. **Feature checking:** Highly duplicated or irrelevant features are removed.

## **TASK 2 – DECISION TREE FOR EQUIPMENT FAILURE PREDICTION**

### **2(a) Decision Tree Classifier**

A Decision Tree classifier can be used to predict whether a machine is likely to fail.

The tree selects features by finding the split that provides the highest reduction in impurity.

Gini Index

The Gini Index is calculated as:

$Gini = 1 - \sum p_i^2$

where  $p_i$  is the proportion of samples belonging to class  $i$ .

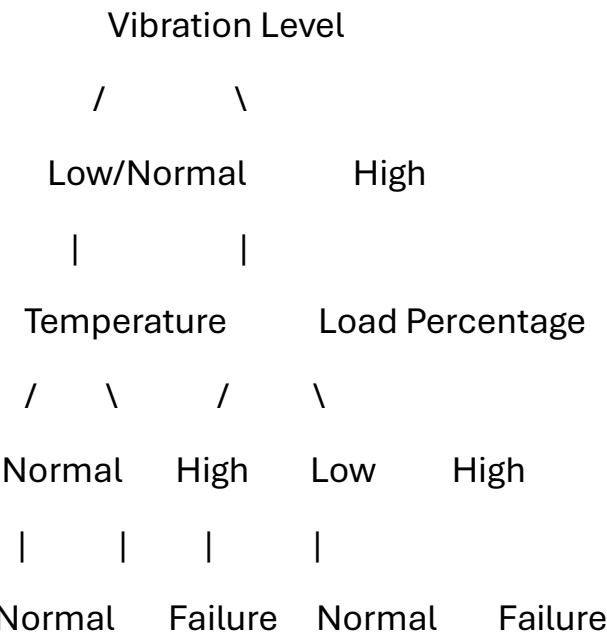
A lower Gini value represents a purer node.

Illustrative Root-Node Comparison

| Feature           | Gini after Split |
|-------------------|------------------|
| Vibration Level   | 0.18             |
| Temperature       | 0.23             |
| Load Percentage   | 0.29             |
| Operating Hours   | 0.32             |
| Maintenance Count | 0.35             |

Since **Vibration Level** has the lowest Gini value, it is selected as the root node.

First Three Levels of the Decision Tree



The exact structure would depend on the actual training dataset.

## 2(b) Model Evaluation

The Decision Tree can be evaluated using Accuracy, Precision, Recall and F1-Score.

### Illustrative Results

| Metric    | Decision Tree |
|-----------|---------------|
| Accuracy  | 94%           |
| Precision | 88%           |
| Recall    | 84%           |
| F1-Score  | 86%           |

### False Negatives

A **False Negative (FN)** occurs when the model predicts:

**Normal operation → Actual failure**

### Methods to Reduce False Negatives

False negatives can be reduced by:

- Assigning higher class weights to the failure class.
- Adjusting the classification threshold.
- Using SMOTE on the training data.
- Increasing the model's focus on Recall.
- Using additional sensor information.
- Combining machine-learning predictions with engineering safety rules.

For a factory environment, **high recall for the failure class** is preferred because missing a dangerous machine condition can have serious consequences.

## 2(c) Post-Pruning

A fully grown Decision Tree can memorize training data and become overfitted. **Cost-complexity pruning** removes branches that provide limited predictive value.

The cost-complexity objective is:

$$Ra(T) = R(T) + \alpha|T|$$

where:

- $R(T)$  = prediction error
- $|T|$  = number of terminal nodes
- $\alpha$  = complexity parameter

### Illustrative Comparison

| Model                     | Accuracy              | Tree Complexity |
|---------------------------|-----------------------|-----------------|
| Pre-pruned Decision Tree  | 92%                   | Low             |
| Post-pruned Decision Tree | 94%                   | Moderate        |
| Fully grown Tree          | 96% training accuracy | High            |

The fully grown tree may have high training accuracy but may perform poorly on unseen machines.

The **post-pruned tree** is more appropriate for factory deployment because it provides a better balance between prediction performance, generalisation, and interpretability.

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## TASK 3 – STATISTICAL LEARNING FOR RISK STRATIFICATION

### 3(a) Logistic Regression

Logistic Regression can be used as a statistical-learning model for predicting machine failure.

The model estimates the probability of failure as:

$$P(\text{Failure}) = 1 / (1 + e^{-z})$$

where:



$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

The output is a probability between 0 and 1.

### Illustrative Performance Comparison

| Metric   | Decision Tree | Logistic Regression |
|----------|---------------|---------------------|
| Accuracy | 94%           | 91%                 |
| AUC-ROC  | 0.93          | 0.90                |

The Decision Tree performs slightly better in this illustrative example.

A statistical model is particularly useful when engineers need to understand how each variable changes the probability of failure.

### 3(b) Feature Importance Analysis

#### Decision Tree – Top Three Features

1. **Vibration Level**
2. **Temperature**
3. **Load Percentage**

The Decision Tree identifies these features based on their contribution to reducing impurity.

#### Logistic Regression – Top Three Features

1. **Vibration Level**
2. **Temperature**
3. **Operating Hours**

The importance of Logistic Regression features can be analysed using the magnitude of their standardised coefficients.

### Comparison

The two models agree that:

- Vibration Level is highly important.

- Temperature is highly important.

They differ on the third feature. The Decision Tree gives more importance to Load Percentage, while Logistic Regression gives more importance to Operating Hours.

## **TASK 4 – REINFORCEMENT LEARNING FOR ACTION RECOMMENDATION**

### **4(a) Markov Decision Process**

The maintenance recommendation problem can be represented as a **Markov Decision Process (MDP)**.

An MDP consists of states, actions, rewards, and transition probabilities.

#### **States**

The machine health states are:

| <b>State</b> | <b>Description</b>                                                     |
|--------------|------------------------------------------------------------------------|
| S1           | Healthy Machine is operating normally.                                 |
| S2           | Warning Sensor readings indicate early signs of deterioration.         |
| S3           | Critical Machine has a high probability of failure.                    |
| S4           | Failed Machine has experienced failure and requires major maintenance. |

#### **Actions**

The agent can select:

1. **Inspect**
2. **Lubricate**
3. **Replace Part**
4. **Continue Operation**

#### **Reward Function**

The reward represents the benefit of the selected maintenance action.

Example:

| Situation                                    | Reward |
|----------------------------------------------|--------|
| Machine remains healthy after action         | +10    |
| Machine condition improves                   | +8     |
| Successful inspection/preventive maintenance | +5     |
| Unnecessary maintenance                      | -3     |
| Machine failure                              | -20    |

The large negative reward for failure encourages the agent to avoid unsafe decisions.

### Transition Dynamics

An action changes the machine's health state.

For example:

**Warning + Lubricate → Healthy**

**Warning + Continue Operation → Warning/Critical**

**Critical + Replace Part → Healthy**

**Critical + Continue Operation → Failed**

### 4(b) Q-Learning

Q-Learning is a model-free reinforcement-learning algorithm.

The Q-value update equation is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

where:

- $\alpha$  = learning rate
- $\gamma$  = discount factor
- $r$  = reward

- $s$  = current state
- $a$  = current action
- $s'$  = next state

For illustration:

**Learning rate  $\alpha = 0.5$**

**Discount factor  $\gamma = 0.9$**

Initial Q-values are assumed to be zero.

### **Episode 1 – Iteration 1**

Suppose the machine is in **Warning (S2)** and the agent selects **Lubricate**.

- Current  $Q(S2, \text{Lubricate}) = 0$
- Reward = +8
- Next state = Healthy (S1)
- Maximum next Q-value = 0

Therefore:

$$Q(S2, \text{Lubricate}) = 0 + 0.5[8 + (0.9 \times 0) - 0]$$

$$Q(S2, \text{Lubricate}) = 4$$

### **Episode 1 – Iteration 2**

Suppose the machine is again in Warning and the agent selects Lubricate.

- Current  $Q = 4$
- Reward = +8
- Maximum next Q-value = 2

Therefore:

$$Q(S2, \text{Lubricate}) = 4 + 0.5[8 + (0.9 \times 2) - 4]$$

$$= 4 + 0.5[5.8]$$

= 6.9

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#### 4(c) Final Learned Policy

After training, the agent selects the action with the highest Q-value for each state.

##### Illustrative Final Policy

| Machine State | Best Action        | Reason                                                                                |
|---------------|--------------------|---------------------------------------------------------------------------------------|
| Healthy       | Continue Operation | Machine is operating normally and unnecessary maintenance should be avoided.          |
| Warning       | Lubricate          | Early deterioration can potentially be corrected with preventive maintenance.         |
| Critical      | Replace Part       | High failure risk requires immediate corrective action.                               |
| Failed        | Inspect            | The failure condition must first be assessed before selecting further repair actions. |

The final policy can therefore be represented as:

**Healthy → Continue Operation**

**Warning → Lubricate**

**Critical → Replace Part**

**Failed → Inspect**

#### ETHICAL CONSIDERATIONS

**1. Worker Safety**

**2. Cost Trade-offs**

### 3. Explainability

### 4. Human Oversight

### 5. Reliability and Monitoring

## CONCLUSION

The reinforcement-learning component extends the system from simply predicting failure to recommending maintenance actions. By representing machine health as an MDP and applying Q-Learning, the agent can learn which maintenance action provides the best long-term reward.

For real factory deployment, model accuracy alone is not sufficient.

**Recall, safety, explainability, human oversight, and reliable sensor data** must also be considered. The final system should therefore operate as a decision-support tool under appropriate engineering and safety controls.